Results

- **FINAL TX/RX**

```
Transmitting:  ID = 0x123 (00010010011)
             DLC = 4 (0100)
            DATA = 0xDEADBEEF

--------------------------------------------------------------------------------
TX FRAME (83 bits):
--------------------------------------------------------------------------------
Field:   SOF ID[10:0]      RTR IDE r0 DLC   DATA0      DATA1      DATA2      DATA3      CRC[14:0]       D ACK D EOF       IFS
TX Bits: 1 10001001000 110 0001 00110111 10101011 01101111 10011101 111100011100110 101 1111011 111
--------------------------------------------------------------------------------
```

```
RX FRAME (from register file - loopback):
--------------------------------------------------------------------------------
  ID [10:0] = 00100100011 (0x123)
  DLC [3:0] = 0100 (4 bytes)
  DATA[0]   = 11011110 (0xde)
  DATA[1]   = 10101101 (0xad)
  DATA[2]   = 10111110 (0xbe)
  DATA[3]   = 11101111 (0xef)
--------------------------------------------------------------------------------

[PASS] TX and RX frames match!
```

- **Block Wise Self Checking Results**

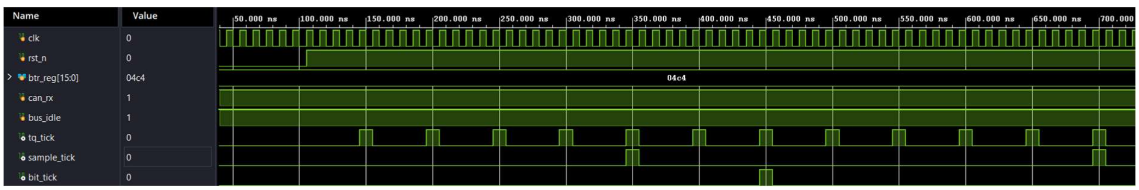
1. **TX FSM**

```
ID=0x123, DLC=1, DATA=0xAB)
  FULL : 0001001000110000001101010111010101010101011111111111111
--------------------------------------------------------------------------------
  SOF         : 0
  ID[10:0]    : 00100100011
  RTR         : 0
  IDE         : 0
  r0          : 0
  DLC[3:0]    : 0001
  DATA byte0  : 10101011
  CRC[14:0]   : 101010101010101
  CRC delim   : 1
  ACK slot    : 1
  ACK delim   : 1
  EOF         : 1111111
  IFS         : 111
```

2. **RX FSM**

```
rx_id      : 0x123
rx_dlc     : 1
rx_data0   : 0x0000 (byte0=00 byte1=00)
rx_data1   : 0x0000
rx_data2   : 0x0000
rx_data3   : 0x00ab
rx_ready   : 0
crc_err    : 0
form_err   : 0
```

3. **BIT TIMING**



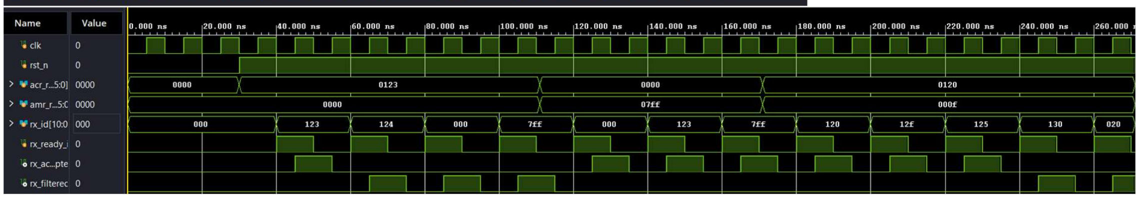

```
# run 10000ns
[PASS] Reset: outputs low
[PASS] tq_tick rate (69)
[PASS] bit_tick rate (9)
[PASS] One sample per bit (10 samples, 9 bits)
[PASS] Hard sync fires
[PASS] No hard sync when bus active
[PASS] New BTR bit rate (5)
```

4. **CRC**

```
[PASS] Reset: crc=0x0000
[PASS] single 1 bit : crc=0x4599
[PASS] pattern 1010 : crc=0x3aac
[PASS] pattern 10110010 : crc=0x1a01
[PASS] eleven zeros : crc=0x0000
[PASS] fifteen ones : crc=0x6806
[PASS]  ID=0x123 D=0xAB : crc=0x666f
```

5. ACCEPTANCE FILTER

```
[PASS] exact match hit : ID=0x123 ACCEPTED
[PASS] exact match miss : ID=0x124 REJECTED
[PASS] exact match zero : ID=0x000 REJECTED
[PASS] t match all-ones : ID=0x7ff REJECTED
```



6. BIT STUFFING

```
TX original input : 0000011
TX stuffed output : 00000111    (extra bit = inserted stuff bit)
RX recovered bits : 0000011
```

7. ARBITRATION

```
[PASS]  (both dominant) : tx=0 rx=0 lost=0
[PASS] (both recessive) : tx=1 rx=1 lost=0
[PASS] cessive than tx) : tx=0 rx=1 lost=0
[PASS] e, bus dominant) : tx=1 rx=0 lost=1
[PASS] tx_active=0: no arb_lost
```

8. ERROR DETECTOR

```
[PASS] Initial: error active, TEC=0 REC=0
[PASS] TX error: TEC=8
[PASS] RX error: REC=1
[PASS] TX success: TEC=7
[PASS] RX success: REC=0
[PASS] Error passive: TEC=135
[FAIL] Bus-off: TEC=7 bus_off=1
[PASS] Bus-off recovery: TEC=0 REC=0
```

9. INTERRUPT

```
[PASS] Initial: error active, TEC=0 REC=0
[PASS] TX error: TEC=8
[PASS] RX error: REC=1
[PASS] TX success: TEC=7
[PASS] RX success: REC=0
[PASS] Error passive: TEC=135
[FAIL] Bus-off: TEC=7 bus_off=1
[PASS] Bus-off recovery: TEC=0 REC=0
```